

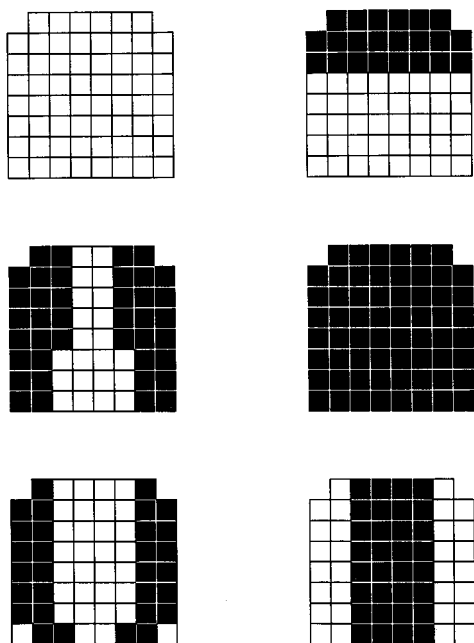


FIG. 2. Linguopalatal contact patterns used in the experiment differing in amount and distribution of electrode activation. (Top left) 0 “on” electrodes, (top right) 22 “on” electrodes as for an alveolar stop; (middle) 40 and 62 “on” electrodes as for a palatal fricative and for an alveopalatal stop respectively; (bottom) 30 “on” electrodes at the palate sides and 32 “on” electrodes at the palate center.

average was subtracted from the measurement value with palate. The same operation was repeated two more times. Data for each subtraction operation were averaged across all temporal frames of each sweep and the resulting averages were averaged among themselves.

Second, two sets of three 5-s sweeps were recorded with the magazine parallel to an artificial palate with 62 activated

electrodes; two distance conditions, i.e., 6–7 mm and 4–5 cm, were tested in this case. In a real speech production setting, the tongue coils touch the palate surface and thus the linguopalatal distance is smaller than 6–7 mm; this distance condition was not tested in the present experiment since a mold was needed to create contact activation on the artificial palate. It is hypothesized that the two unrealistic distance conditions tested here should provide information about possible effects in a real speech production setting, provided they yielded analogous coil distortion values.

The average data were subtracted from the values obtained with the artificial palate subject to no electrode activation (i.e., with 0 “on” electrodes). The reason for choosing a nonactivated palate instead of no palate as reference condition was the difficulty of pulling out the artificial palate from the magazine without affecting the position of the coils when the former was parallel to the latter.

The data were obtained in a similar fashion to the perpendicular case: Each relevant measurement was preceded and followed by a reference measurement taken with the artificial palate subject to no contact activation (i.e., with 0 “on” electrodes). The preceding and following reference measurements were averaged and subtracted from the relevant measurement. This operation was repeated three times. The three resulting values were averaged among themselves following the procedure for the perpendicular condition.

In order to find out whether the transducer coils would interfere with the linguopalatal contact pattern, the second author of this paper read a list of sequences with an artificial palate in place both with and without three coils attached to the midline of the tip, blade, and dorsum. The fast setting glass ionomer cement ESPE Ketac Bond was used for attaching the coils to the tongue surface (the cement extended about 1 mm around the coils). The tongue tip coil was placed

TABLE I. Effects on coil position from differences in contact size on an artificial palate placed perpendicularly to the magazine. They are reported as a function of coil position number (1–9; see Fig. 1) and contact size (0, 22, 40, and 62 electrode activation; see Fig. 2, top and middle); effects for the mold condition alone are also given. Values (in mm) represent differences in coil position between the mold and contact size conditions, on the one hand, and a condition involving no artificial palate or mold, on the other hand. They were averaged across sweeps and artificial palates for the x and y components (within parentheses) and for the Euclidean distance r (below).

	Mold	0 electrodes	22 electrodes	40 electrodes	62 electrodes
1	(0,0.01) 0.01	(0,0) 0	(−0.04,0.04) 0.06	(−0.07,0.06) 0.09	(−0.03,0.08) 0.08
2	(−0.01,0) 0.01	(−0.02,0.02) 0.03	(−0.04,0.07) 0.08	(−0.04,0.05) 0.06	(−0.04,0.08) 0.09
3	(−0.02,0.02) 0.03	(0,0.06) 0.06	(−0.08,0.07) 0.11	(−0.11,0.07) 0.13	(−0.12,0.13) 0.18
4	(0,0) 0	(−0.06,−0.04) 0.07	(−0.08,0.09) 0.12	(−0.13,0.13) 0.18	(−0.15,0.15) 0.21
5	(−0.05,0.02) 0.05	(−0.05,0.03) 0.06	(−0.10,0.09) 0.13	(−0.17,0.10) 0.20	(−0.16,0.11) 0.19
6	(−0.04,−0.01) 0.04	(−0.03,0.05) 0.06	(−0.09,0.08) 0.12	(−0.08,0.09) 0.12	(−0.10,0.12) 0.16
7	(−0.05,0.05) 0.07	(−0.01,0.06) 0.06	(−0.07,0.08) 0.11	(−0.10,0.08) 0.13	(−0.12,0.10) 0.16
8	(−0.03,0.04) 0.05	(−0.03,0.07) 0.08	(−0.05,0.10) 0.11	(−0.05,0.08) 0.09	(−0.07,0.10) 0.12
9	(0.04,−0.01) 0.04	(0,0.07) 0.07	(−0.03,0.06) 0.07	(−0.04,0.09) 0.10	(−0.02,0.09) 0.09